What a 28-Subject Test Set Can and Cannot Tell You: Distillation, Failure Modes, and Pipeline Defects in Forensic Bruise Segmentation

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Abstract

Photographic documentation of bruising is central to forensic evidence, and an automated segmenter for point-of-care use must be accurate enough to document an injury while small enough to run on modest hardware. We build such a pipeline—knowledge distillation from a SegFormer-B2 teacher into deployment-sized students, plus YOLO26n-sem baselines—and evaluate it on a locked, subject-disjoint test set of 185 images. Our main contribution is not the pipeline but an honest accounting of what that test set can support. The 185 images come from only 28 subjects, and images of one subject are not independent; under a *paired, subject-level cluster bootstrap*, none of the accuracy differences one would normally headline survives. Distillation improves median Dice by $+0.018$, but with 95% CI $[-0.004, +0.044]$. The distilled student matches its teacher ($+0.001$, CI $[-0.027, +0.042]$) at $7.4\times$ fewer parameters, which is what distillation promises—but "matches" here means "indistinguishable at this sample size," not "equivalent." What *is* resolvable is an order of magnitude larger and lives elsewhere. First, failure modes: YOLO26n-sem attains the best median Dice of any model (0.849) yet returns a completely empty mask on 9.7% of images containing a bruise, $+9.19$ percentage points more than the distilled student (CI $[+1.83, +17.54]$, significant), and distillation significantly reduces YOLO's miss rate (-2.70 pp, CI $[-5.03, -0.80]$). Second, pipeline defects: evaluating YOLO's raw head through an ImageNet-normalised dataloader, rather than the $[0, 1]$-scaled letterboxed input it was trained on, costs 0.230 mean Dice (CI $[+0.112, +0.344]$) and doubles the miss rate—with identical weights and an identical scoring grid, and with no exception, no warning, and an entirely plausible number as the only symptom. We conclude that at realistic clinical sample sizes, benchmark tables reporting bare Dice point estimates are largely reporting noise, while catastrophic failure rates and silent pipeline bugs are both detectable and consequential.

1 Introduction

Bruising is among the most common physical findings in assault, intimate partner violence, and child maltreatment casework, and photographs of bruises are routinely entered as evidence. Automating their delineation is attractive for consistency and throughput, but the deployment context is unusually constrained: examinations happen at bedside and in clinics, not next to a datacentre, so a model must be small and fast; and the artifact of interest is the documented *extent and shape* of the injury overlaid on the original photograph, so classification is insufficient and segmentation is required.

Knowledge distillation [Hinton et al., 2015] addresses the size/accuracy tension directly: a high-capacity teacher supplies soft, graded supervision that a small student can absorb, ideally retaining teacher-level accuracy at student-level cost. For dense prediction the motivation is stronger than for classification, because a boundary pixel's label is genuinely uncertain and a teacher probability of 0.58 communicates that where a hard 1 cannot.

This paper reports three findings, and they are ordered by how much we think they should be believed.

At 28 subjects, no Dice difference we care about is resolvable. The obvious result to report is that our distilled SegFormer-B0 student reaches median Dice 0.828 against 0.811 for an identically-architected non-distilled baseline and 0.827 for its own $7.4\times$ larger teacher. All three numbers are real. None of the differences between them survives a paired subject-level cluster bootstrap: distillation gives $+0.018$ median Dice with CI $[-0.004, +0.044]$, and the student–teacher difference is $+0.001$ with CI $[-0.027, +0.042]$ (Table 2). The 185 test images come from only 28 subjects, and images of a subject share skin, lighting, and injury; the effective sample size is 28, not 185. We report the point estimates because they are the honest output of the experiment, and we report the intervals because they are what determines whether the point estimates mean anything. The practical reading is that our student *matches* its teacher at $7.4\times$ fewer parameters and roughly twice the throughput—which is what distillation is for—but that this test set cannot distinguish it from the non-distilled baseline either.

Failure modes are resolvable, and they invert the ranking. YOLO26n-sem [Jocher et al., 2026] is the smallest and fastest model we train and attains the highest median Dice of any (0.849). It also returns an entirely empty mask on

9.7% of images that contain a bruise, versus 0.5% for the distilled student—a difference of $+9.19$ percentage points with CI $[+1.83, +17.54]$, which *is* significant, unlike any of the Dice differences above. Distillation likewise cuts YOLO's miss rate significantly (-2.70 pp, CI $[-5.03, -0.80]$) while changing its Dice not at all. These facts are consistent: the model is excellent on typical images and brittle on hard ones, which is exactly what a median rewards and a miss rate exposes. In a forensic setting the asymmetry is decisive—a loose boundary yields a document an examiner can inspect and correct, whereas a blank mask yields silence about an injury that is present—so we recommend the distilled SegFormer despite YOLO's better headline score. The general point is sharper than the recommendation: the quantity that ranks these models is the one this sample size can actually measure, and it is not Dice.

Leaving a framework's native inference path is not free, and fails silently. To obtain threshold and temperature control that Ultralytics' `argmax` postprocessing does not expose, we evaluated YOLO's raw semantic head directly. The move is algebraically sound—two-class `argmax` is exactly thresholding $\sigma(z_1 - z_0)$ at 0.5—but it silently inherits an obligation to reproduce the framework's preprocessing. Ours did not: it applied ImageNet normalisation and a stretch-resize to weights trained on $[0, 1]$-scaled, letterboxed input. Holding the checkpoint and the scoring grid fixed, this costs 0.230 mean Dice (subject-level bootstrap 95% CI $[0.112, 0.344]$) and roughly doubles the complete-miss rate. Nothing raises an error; the pipeline simply reports a lower, entirely plausible number that no downstream threshold or temperature search can recover, because the corruption is upstream of the logits being calibrated. We recommend that any evaluation departing from a framework's native path be accompanied by a positive control reproducing that framework's native performance.

2 Related Work

Efficient semantic segmentation. Following the Vision Transformer [Dosovitskiy et al., 2021] and the attention mechanism it builds on [Vaswani et al., 2017], SegFormer [Xie et al., 2021] pairs a hierarchical Mix Transformer (MiT) encoder with a deliberately lightweight all-MLP decoder and scales from B0 to B5 within one design. The B0/B2 pair is therefore a natural teacher–student couple: identical inductive biases and output interface, differing only in capacity, which isolates the effect of distillation from any architectural confound. The YOLO family [Redmon et al., 2016] targets the opposite end of the latency budget; recent versions [Jocher et al., 2026] add a first-class semantic segmentation head, making a single-stage real-time detector usable for dense prediction.

Knowledge distillation. Hinton et al. [2015] introduced training a student against a teacher's softened output distribution, the soft targets carrying information about relative confidence that hard labels discard. Feature-level variants [Romero et al., 2015] match intermediate representations rather than outputs.

Calibration. Modern networks are systematically overconfident [Guo et al., 2017]. Temperature scaling—dividing logits by a scalar fitted on held-out data—is the standard single-parameter remedy, and is what makes a saturated teacher's probability map informative enough to distil from.

Bruise imaging. Bruise visibility depends strongly on imaging modality and on skin pigmentation: in a randomised controlled trial spanning six skin-colour categories, Scafide et al. [2020] found alternate light roughly five times more likely than white light to reveal bruising, with follow-up work examining visibility over time [Downing et al., 2024]. We work in the white-light regime, which is what is available in the majority of examination settings.

3 Problem Setting and Data

Task. Given a white-light RGB photograph, predict a binary mask $\hat{Y} \in \{0, 1\}^{H \times W}$ with $\hat{Y}_{ij} = 1$ where pixel (i, j) is bruised.

Dataset. We use 1,016 white-light photographs at native 4022×6024 resolution with consensus-annotated bruise masks, split into 697 training, 134 validation, and 185 fixed-test images.

Leakage control. Because multiple photographs exist per subject and per visit, image-level splitting would leak subject-specific skin appearance between train and test. Splits are therefore *subject-level disjoint*: the 115 subjects in train/validation and the 28 subjects in the test set share no individual. We verify this programmatically rather than by construction alone; the measured intersection is empty.

Protocol discipline. The test set is locked. Thresholds, distillation weights, and model selection are chosen exclusively on validation; the test set is read once per model, after all such choices are frozen.

4 Method

4.1 Architectures

SegFormer. An input $x \in \mathbb{R}^{B \times 3 \times 640 \times 640}$ passes through the MiT encoder, which emits features at four scales ($H/4$, $H/8$, $H/16$, $H/32$). Overlapping patch embeddings preserve local continuity across patch borders, which matters here because bruise boundaries are diffuse rather than sharp and non-overlapping tokenisation tends to produce blocky edges. Each block applies

$$X' = X + \mathrm{Attention}(\mathrm{LN}(X)), \tag{1}$$
$$X'' = X' + \mathrm{MixFFN}(\mathrm{LN}(X')). \tag{2}$$

Full self-attention over $N = HW$ tokens costs $O(N^2)$, prohibitive at segmentation resolution. SegFormer applies sequence reduction with ratio R: keys and values are reshaped and projected from N to N/R entries before

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_{\text{head}}}}\right) V, \tag{3}$$

reducing cost to $O(N^2/R)$ while queries retain full spatial resolution. Mix-FFN inserts a 3×3 depthwise convolution inside the feed-forward block,

$$\mathrm{MixFFN}(X) = \mathrm{MLP}\big(\mathrm{GELU}(\mathrm{DWConv}(\mathrm{MLP}(X)))\big) + X, \tag{4}$$

supplying positional information implicitly and removing the fixed positional embeddings that would otherwise need interpolating whenever test resolution differs from training resolution—a practical advantage for a system whose input is whatever the camera produces. The all-MLP decoder projects all four stages to a common width, upsamples to $H/4$, concatenates, fuses, and classifies. We replace the multi-class head with a single-channel head, so the model emits one bruise logit per pixel; our wrapper bilinearly upsamples logits to the input grid, giving $z_S \in \mathbb{R}^{B\times1\times640\times640}$.

YOLO26n-sem. A backbone–neck–head detector whose semantic head emits dense two-class logits $Z \in \mathbb{R}^{B\times2\times H\times W}$ (channel 0 background, channel 1 bruise). Ultralytics' native postprocessing converts these to a hard class map by `argmax`. For a two-class head this is equivalent to thresholding the logit difference at zero:

$$\frac{e^{z_1}}{e^{z_0} + e^{z_1}} = \sigma(z_1 - z_0), \tag{5}$$

so $\arg\max$ coincides with $\sigma(z_1 - z_0) \geq 0.5$. This identity is what makes a threshold- and temperature-controlled path *appear* to be a free generalisation of `argmax`. Section 6.3 shows why it is not.

4.2 Supervised Objective

We minimise a sum of binary cross-entropy and a soft Dice term:

$$\mathcal{L}_{\text{sup}} = \mathrm{BCE}(z, Y) + \left(1 - \frac{2\sum_{ij} p_{ij} y_{ij} + s}{\sum_{ij} p_{ij} + \sum_{ij} y_{ij} + s}\right), \tag{6}$$

with $p = \sigma(z)$ and smoothing constant s. The terms are complementary and neither suffices alone. BCE is decomposable over pixels and gives dense, well-conditioned gradients everywhere, but it is dominated by the background class: bruises occupy a small fraction of most photographs, so predicting "background" everywhere already achieves low BCE. The Dice term [Milletari et al., 2016] is a direct differentiable surrogate for the overlap metric we report and is invariant to the foreground/background ratio, which counteracts exactly that imbalance; but alone it is unstable early in training, when the intersection is near zero and its gradient is poorly conditioned. BCE stabilises the start; Dice aligns the objective with the metric.

4.3 Teacher Calibration

A distillation target is useful only insofar as it is graded. A BCE-trained segmentation teacher drives confident pixels toward $\pm\infty$, so its raw probability map is nearly binary and carries little more than the hard mask. We fit a single temperature T on validation by minimising negative log-likelihood [Guo et al., 2017],

$$P_T = \sigma(z_T/T), \tag{7}$$

obtaining $T = 1.840$, which reduces validation NLL from 0.0634 to 0.0452. T is fitted post hoc on held-out data and changes no weights; it exists solely to re-expand saturated logits into the informative region of the sigmoid before they are used as targets.

4.4 Distillation

SegFormer (online). The frozen teacher supplies P_T under `no_grad` each batch, and the student minimises

$$\mathcal{L} = \alpha\,\mathcal{L}_{\text{sup}}(z_S, Y) + (1-\alpha)\,\mathrm{BCE}(z_S, P_T). \tag{8}$$

The student always retains access to ground truth: the teacher is imperfect, and α anchors the student to the consensus annotation while the second term transfers the teacher's boundary uncertainty.

YOLO (offline). Ultralytics' training loop does not expose a per-batch hook for an auxiliary teacher loss, so online distillation in the form of Eq. 8 is unavailable. We instead fuse teacher probability with ground truth and binarise into a pseudo-mask before training:

$$\text{fused} = \alpha Y + (1-\alpha) P_T, \qquad \text{pseudo} = \mathbb{1}[\text{fused} \geq 0.5]. \tag{9}$$

This is a strictly weaker transfer—hard pseudo-labels discard the graded confidence that motivates distillation in the first place—and we flag it as a limitation rather than an equivalent alternative.

4.5 Threshold Selection

SegFormer emits probabilities; a mask requires a decision, $\hat{Y}_{ij} = \mathbb{1}[p_{ij} \geq \tau]$. Because τ trades precision against recall and the optimum is dataset-dependent (there is nothing special about 0.5), we sweep $\tau \in \{0.05, 0.10, \ldots, 0.95\}$ and select the validation-maximising value, then freeze it before touching the test set. YOLO's native path exposes no such knob: `argmax` fixes the operating point at the $\sigma(z_1 - z_0) = 0.5$ boundary by construction.

4.6 Metrics

With TP, FP, FN pixel counts we report $\text{Dice} = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN})$, $\text{IoU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN})$, precision, and recall. We lead with *median* Dice: the per-image distribution is strongly left-skewed by a minority of hard images, so the mean conflates typical-case quality with failure frequency, and we prefer to report those separately. Failure frequency is reported explicitly as the **complete-miss rate**, the fraction of images containing a bruise for which the model predicts zero positive pixels. We separate it from Dice because it is the decisive failure: a loose boundary is a usable document, whereas a blank mask is a missed injury.

All metrics are computed on a common 640×640 grid, with masks resized by nearest-neighbour interpolation, which preserves binary labels where bilinear would introduce fractional ones. Confidence intervals, where reported, come from a bootstrap resampling *subjects* rather than images [Efron and Tibshirani, 1993], since images from one subject are not independent.

5 Experimental Setup

Table 3 lists the configuration; we justify the non-obvious choices here rather than leaving them as magic numbers.

Resolution (640×640). This is YOLO26n-sem's native training resolution, so adopting it avoids handicapping that baseline. It also suits SegFormer's four-stage encoder: 640 is divisible by 32, so every stage stride ($/4, /8, /16, /32$) lands on an integer grid with no padding artifacts.

Learning rates (6×10^{-5} backbone, 6×10^{-4} decoder). We follow Xie et al. [2021]. The asymmetry is deliberate and not a tuning artifact: the MiT encoder is pretrained on ImageNet [Deng et al., 2009] and needs only gentle adaptation—too high a rate erases the pretrained features—whereas the decode head is randomly initialised and must learn from scratch, so it receives $10\times$ the rate to catch up rather than bottleneck the encoder.

Batch size (8 for B2, 32 for B0). Batch size is not hand-set. A VRAM probe raises it per model toward a target occupancy, capped at 32; the 3.71M-parameter B0 fits the cap, while the 27.35M-parameter B2 admits only 8 at 640^2. The consequence is visible in the schedule: with 697 training images, B2 runs 87 optimiser steps per epoch and B0 only 21, so the two students see the same data under materially different step counts. This is a confound we accept rather than hide—equalising batch size would have required either starving B0 or gradient-accumulating B2—and it is one reason we treat the small teacher–student difference as a match rather than a win.

Optimiser and decay. AdamW [Loshchilov and Hutter, 2019] with $\beta = (0.9, 0.999)$ and weight decay 0.01; decay is disabled on norm and bias parameters, where it acts as arbitrary shrinkage on scale parameters rather than capacity control. Learning rate warms up linearly over the first 1% of steps—long enough to avoid the large, badly-directed updates that destabilise a pretrained encoder at step zero, short enough not to waste budget—then decays polynomially with $p = 1.0$ (i.e. linearly) to zero.

Distillation weight α. Rather than hand-set the single most outcome-determining hyperparameter of the method under study, we search it with Optuna [Akiba et al., 2019] (15 trials, 15 epochs each, TPE, range $[0.10, 0.90]$ in steps of 0.05), selecting on validation Dice. This yields $\alpha = 0.60$ for SegFormer-B0 and $\alpha = 0.40$ for YOLO. Both land in the interior of the range, indicating a genuine optimum rather than a boundary artifact, and both weight ground truth and teacher comparably—consistent with the teacher being informative but imperfect.

Other. Gradient clipping at 1.0; mixed precision enabled; early stopping on validation Dice with patience 15 inside a 100-epoch budget (B2 stopped at 30 epochs, B0-direct at 37, B0-distilled at 46); seed 42 throughout. Implementation uses PyTorch [Paszke et al., 2019], HuggingFace Transformers [Wolf et al., 2020], Ultralytics [Jocher et al., 2026], and Albumentations [Buslaev et al., 2020].

6 Results

6.1 Point Estimates, and Why They Are Not Enough

Table 1 reports the fixed-test results with subject-level cluster-bootstrap intervals. Read as point estimates, the story is tidy: SegFormer-B0-distilled improves over the identically-architected B0-direct baseline on median Dice (0.828 vs. 0.811) and mean Dice (0.787 vs. 0.775), and matches its own teacher (0.827) at $7.4\times$ fewer parameters and double the throughput (95.6 vs. 47.6 FPS). Since the two students are the same network on the same data with the same schedule, differing only in the teacher term, any real gap would be attributable to distillation alone.

The intervals do not support the story. Table 2 gives the *paired* bootstrap—both models scored on the same resampled subjects, which is the correct test for same-test-set comparisons and strictly tighter than checking whether the marginal

Table 1: Fixed-test results (185 images from 28 subjects, all metrics on the same 640×640 grid). Brackets are 95% subject-level cluster-bootstrap intervals (B=4000, resampling subjects). SegFormer operating points are validation-selected thresholds; YOLO uses Ultralytics' native `argmax` (no tunable threshold). Throughput is fp32 raw-forward on one A100. *The intervals are wide: at 28 subjects most between-model accuracy differences here are not resolvable* (Table 2).

Model	Training	Params	τ	Med. Dice [95% CI]	Mean Dice [95% CI]	Miss%	FPS
SegFormer-B2 (teacher)	teacher	27.35M	0.75	0.827 [0.771, 0.874]	0.767 [0.713, 0.814]	0.00 [0.00, 0.00]	47.6
SegFormer-B0 (direct)	direct	3.71M	0.60	0.811 [0.765, 0.850]	0.775 [0.747, 0.805]	0.54 [0.00, 1.80]	96.0
SegFormer-B0 (distilled)	distilled	3.71M	0.55	0.828 [0.803, 0.856]	0.787 [0.757, 0.816]	0.54 [0.00, 1.80]	95.6
YOLO26n-sem (direct)	direct	1.63M	argmax	0.849 [0.789, 0.883]	0.735 [0.653, 0.812]	9.73 [2.40, 17.93]	104.3
YOLO26n-sem (distilled)	distilled	1.63M	argmax	0.843 [0.785, 0.877]	0.741 [0.654, 0.818]	7.03 [0.56, 14.91]	101.7

Table 2: Paired subject-level cluster bootstrap (B=4000) for each comparison the paper makes. Both models are evaluated on the same 185 images, so differences are paired; each bootstrap iteration resamples subjects once and scores both models on that resample. Intervals excluding zero are marked **sig**. Only the *failure-mode* differences are resolvable at this sample size—no Dice difference is.

Comparison	Δ Med. Dice [95% CI]	Δ Mean Dice [95% CI]	Δ Miss% [95% CI]
Distillation: B0-distilled − B0-direct	+0.018 [-0.004, +0.044]	+0.012 [-0.006, +0.032]	+0.00 [+0.00, +0.00]
Student − teacher: B0-distilled − B2	+0.001 [-0.027, +0.042]	+0.021 [-0.016, +0.060]	+0.54 [+0.00, +1.80]
YOLO-direct − B0-distilled	+0.020 [-0.025, +0.041]	-0.052 [-0.118, +0.010]	**+9.19 [+1.83, +17.54]** sig
YOLO distillation: distilled − direct	-0.005 [-0.032, +0.013]	+0.006 [-0.019, +0.030]	**-2.70 [-5.03, -0.80]** sig

Table 3: Training configuration. SegFormer follows the encoder/decoder learning-rate recipe of Xie et al. [2021]; YOLO follows Ultralytics' own recipe [Jocher et al., 2026]. Batch size is not hand-set: a VRAM probe raises it per model up to a cap of 32, which B0 fits and the larger B2 does not. Distillation weights α were selected by Optuna.

Setting	SegFormer	YOLO26n-sem
Resolution	640×640	
Optimiser	AdamW	auto (Ultralytics)
Backbone LR	6×10^{-5}	—
Decoder LR	6×10^{-4} ($10\times$)	—
Weight decay	0.01	0.0005
LR schedule	poly, p=1.0	cosine, lr_f=0.01
Warmup	1% of steps	3 epochs
Batch (effective)	8 (B2) / 32 (B0)	32
Steps/epoch	87 (B2) / 21 (B0)	—
Max epochs / patience	100 / 15	
Grad. clip	1.0	—
AMP	enabled	
Seed	42	
Teacher temperature T	1.840 (NLL-fitted on val)	
Distillation α	0.60 (Optuna)	0.40 (Optuna)

intervals in Table 1 overlap. Even so, the distillation effect is $+0.018$ median Dice with CI $[-0.004, +0.044]$ and $+0.012$ mean Dice with CI $[-0.006, +0.032]$: both intervals contain zero. The student–teacher difference is likewise indistinguishable from zero.

We want to be precise about what this does and does not say. It does *not* say distillation fails; the point estimates are positive on every Dice metric, and the experiment is simply underpowered to resolve an effect of this size at $n = 28$ subjects. Nor does "matches the teacher" establish

Table 4: Evaluation-protocol ablation. *Identical checkpoints and identical* 640×640 *scoring grid in every row*; only the input pipeline changes. The custom path applies ImageNet normalisation and a stretch-resize; the native path applies Ultralytics' $[0, 1]$ scaling and letterbox, matching how the weights were trained. Bypassing a framework's native preprocessing to gain threshold control is not free.

Model	Input pipeline	Mean Dice	Med. Dice	Miss
YOLO-direct	custom (ImageNet norm)	0.506	0.647	19.4
	native (Ultralytics)	0.735	0.849	9.7
YOLO-distilled	custom (ImageNet norm)	0.608	0.726	9.7
	native (Ultralytics)	0.741	0.843	7.0

equivalence—failing to reject a difference is not evidence of its absence, and with a CI half-width of roughly 0.035 we could not have detected a genuine 2-point Dice deficit in the student had one existed. What it does say is that a table of bare Dice point estimates from a test set this size— the standard artifact of this literature—does not licence the conclusions routinely drawn from it.

6.2 Aggregate Rank and Deployment Fitness Disagree

YOLO26n-sem-direct attains the best median Dice of any model (0.849) while being the smallest (1.63M parameters) and fastest (104.3 FPS). Taken at face value, it wins. Its complete-miss rate, however, is 9.7%: on roughly one image in ten containing a bruise, it returns an empty mask. The distilled SegFormer-B0 misses 0.5%, and the teacher 0.0%.

Crucially, this is the one model-vs-model comparison that the sample size can resolve. The paired difference is $+9.19$

percentage points with CI $[+1.83, +17.54]$ (Table 2), excluding zero, while the same models' median-Dice difference ($+0.020$, CI $[-0.025, +0.041]$) does not. The same holds within YOLO: distillation cuts the miss rate by 2.70 pp (CI $[-5.03, -0.80]$, significant) while moving Dice not at all—consistent with the teacher term suppressing precisely the confident-empty failure mode rather than sharpening boundaries.

This is the paper's central empirical observation. The metric that separates these models at achievable sample sizes is the failure rate, not the overlap score, and the two disagree about which model to deploy. A benchmark reporting only Dice would have ranked YOLO first, reported a difference it could not actually resolve, and hidden the 9.7% of cases where the tool silently returns nothing.

6.3 The Cost of Leaving the Native Inference Path

Our custom evaluation path bypassed Ultralytics' postprocessing to expose threshold and temperature control, justified by the identity in Eq. 5. The identity is correct; the implementation carried an unstated obligation. It fed the raw semantic head through a generic ImageNet-normalised dataloader with a stretch-resize, whereas the checkpoint had been trained through Ultralytics' pipeline, which scales to $[0, 1]$ and letterboxes. The mismatch is silent—no exception, no warning, just a different input distribution reaching filters that never learned it.

Table 4 isolates the effect. Checkpoints are identical, the scoring grid is identical (640×640), and only the input pipeline differs. Mean Dice for YOLO-direct falls from 0.735 to 0.506, a paired difference of 0.230 (subject-level bootstrap 95% CI $[0.112, 0.344]$), and the complete-miss rate doubles from 9.7% to 19.5%. The effect reproduces on the distilled model ($0.741 \rightarrow 0.608$; paired difference 0.133, CI $[0.037, 0.223]$).

Two points deserve emphasis. First, no amount of threshold or temperature search recovers this, because the corruption is upstream of the logits being calibrated: one is tuning an operating point on top of an already-degraded signal. Indeed the calibration search actively misleads—it returns a confident "optimal" (τ, T) pair for a distribution the model was never trained on. Second, and more general: the failure mode of this class of bug is not a crash but a *plausible number*. Had we not compared against the framework's own postprocessing, we would simply have concluded that YOLO is a weaker architecture for this task and reported it as such. We therefore recommend a positive control—reproduce the framework's native performance through your own path before trusting anything that path produces—as cheap insurance whenever an evaluation departs from a vendor's inference code.

7 Discussion and Limitations

On statistical power. The honest summary of our accuracy results is that the experiment is underpowered for the effects it was built to measure. With 28 subjects and a paired cluster bootstrap, the smallest median-Dice difference we could reliably detect is roughly 0.04—larger than the distillation effect (0.018), the student–teacher gap (0.001), and the SegFormer–YOLO gap (0.020) alike. This is not a defect peculiar to our dataset. Forensic and clinical test sets are subject-limited by construction: recruiting more photographs of the same person adds images but very little independent information, and the field's convention of reporting per-image n invites exactly this over-statement. We would rather report a wide interval than a confident point estimate the data cannot sustain, and we suggest that subject-level intervals be reported as a default in this literature rather than as a robustness appendix.

Other limitations. (i) YOLO's distillation is offline pseudo-mask fusion, a strictly weaker mechanism than the online soft-target transfer used for SegFormer; the YOLO direct-vs-distilled comparison is therefore not a clean test of distillation, and its (significant) miss-rate improvement may understate what online transfer would achieve. (ii) Every number is a single training run; we report no seed variance, which compounds the power problem above—the intervals here capture test-set sampling only, not training stochasticity, and are therefore optimistic. (iii) B2 and B0 train at different batch sizes (8 vs. 32) and hence different step counts, an unequalised confound in the teacher–student comparison. (iv) The test set is from a single site. (v) Throughput is raw-forward on one A100 and excludes preprocessing and mask upsampling, so it upper-bounds deployable frame rate. (vi) We do not report a skin-tone fairness analysis. Bruise visibility varies with pigmentation [Scafide et al., 2020], and we do stratify by Individual Typology Angle [Chardon et al., 1991]; but the standard max–min subgroup gap is an extremum-of-extrema statistic, and at 9–17 subjects per group its subject-level bootstrap interval spans roughly $[0.04, 0.89]$—wide enough to be consistent with almost any claim. Reporting it as a point estimate, as is conventional, would be indefensible on this sample. We regard the correct treatment of subgroup-gap uncertainty at realistic clinical sample sizes as the most important open problem this work surfaced.

8 Conclusion

We set out to build a deployable distilled bruise segmenter and largely succeeded: a SegFormer-B0 student matches its B2 teacher at $7.4\times$ fewer parameters and twice the throughput. But the more useful result is what the evaluation could not establish. Under a paired subject-level bootstrap, none of the Dice differences that such papers normally headline—distillation over baseline, student versus teacher, transformer versus detector—is distinguishable from zero at 28 subjects, even though every point estimate points the expected way.

Two things were resolvable, and neither is an overlap score. The first is failure rate: YOLO26n-sem posts the best median Dice of any model while silently returning an empty mask on 9.7% of images with a bruise, $+9.19$ points more than the distilled student (CI $[+1.83, +17.54]$), and this—not Dice—is what determines which model should be deployed. The second is pipeline integrity: evaluating YOLO's head outside its native inference path cost 0.230 mean Dice (CI $[+0.112, +0.344]$) and doubled the miss rate, announcing itself with nothing but a plausible-looking number.

The pattern is that the effects a subject-limited test set can actually measure are an order of magnitude larger than the effects we usually write about, and they concern how a model fails and whether the harness is correct, rather than where it lands on a leaderboard. We suggest three cheap defaults: report subject-level intervals rather than per-image ones; report an explicit catastrophic-failure rate alongside overlap; and, whenever an evaluation leaves a vendor's inference path, require it to reproduce the vendor's own result as a positive control before believing anything else it produces.

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